MODULE	**SCELSE Summer Course 2026**
Instructor	Dr Ezequiel Santillan (esantillan@ntu.edu.sg)
Topics	• Experimental Design • Data Analysis Hands-on Session
Syllabus	<u>Experimental Design</u>: Logic; observations, models, and hypotheses; variability; sampling; replication; independence; temporal and spatial confounding; scale; controls; correlation versus causation; batch effects. <u>Data Analysis Hands-on session</u>: Univariate and multivariate statistics, continuous and categorical covariates, linear models, test statistics, assumptions, p-values, fixed and random factors, crossed vs. nested factors, distance-based multivariate statistics. Basic concepts of experimental design and techniques and tools for data visualization in biology.
Learning objectives	<u>Experimental Design</u>: ○ Understand the logic behind representative sampling and rigorous experimental design. ○ Apply experimental design principles to ecological experiments, including the use of controls. ○ Understand the importance of defining the hypothesis of interest before sampling or conducting experiments. ○ Understand the importance of replication at the appropriate level and ensuring independence among observations. <u>Data Analysis Hands-on session</u>: - Understand and perform statistical analyses of simple univariate datasets. - Understand and perform basic distance-based multivariate analyses. - Identify key principles for effective data visualization. - Recognize resources for further learning.
Contact hours	1 hour (23 June 2026), 1 hour (24 June 2026), 2 hours (25 June 2026)
Interactive activities	<u>Experimental Design</u>: Class discussion <u>Data Analysis Hands-on session</u>: Data Analysis Hands-on session: Class discussion; students will work on data analysis problems to reinforce concepts taught in the *Experimental Design* module.
Related Topics	Of general importance in all subject areas
Suggested Readings	- *Practical Statistics for Data Scientists*, by A. Bruce, P. C. Bruce, and P. Gedeck (O'Reilly 2018) - *R for Data Science: Import, Tidy, Transform, Visualize, and Model Data* by H. Wickham, M. Cetinkaya-R., and G. Grolemund (O'Reilly 2023) - *Modern Statistics for Modern Biology*, by Susan Holmes and Wolfgang Huber (Cambridge University Press 2020) - *Making Data Visual*, by D. Fisher and M. Meyer (O'Reilly 2018) *Nature Collections: Visual Strategies for Biological Data* (Springer Nature, 2015) - *Experiments in Ecology*, by A.J. Underwood (Cambridge University Press 1997, 2012). - *Statistics (4th ed.)*, by D. Freedman, R. Pisani, and R. Purves (W. W. Norton & Company, 2012)